

Individual Assignment 5

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1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

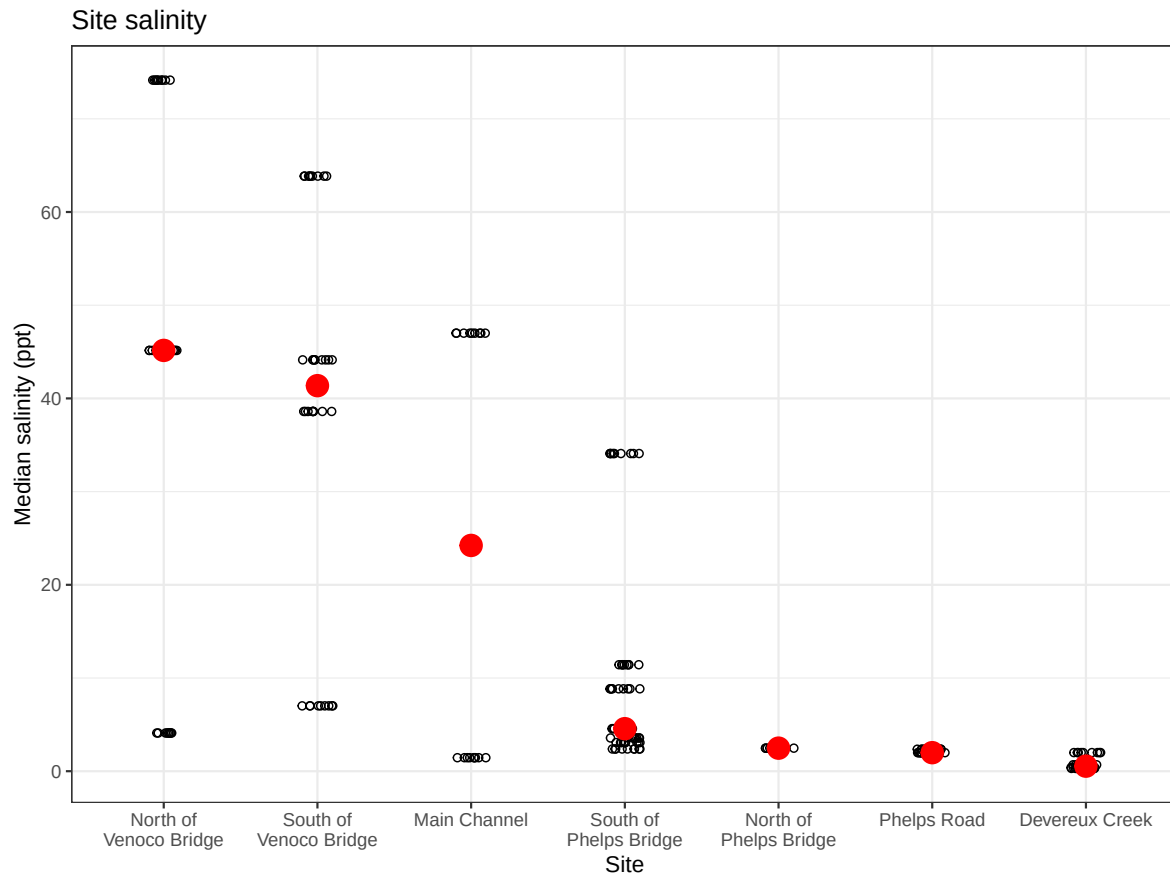
Importing other graphs

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
  color = "red",
  size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
  y = "Median salinity (ppt)",
  title = "Site salinity") +
# different theme
theme_bw()

# displaying object
salinity

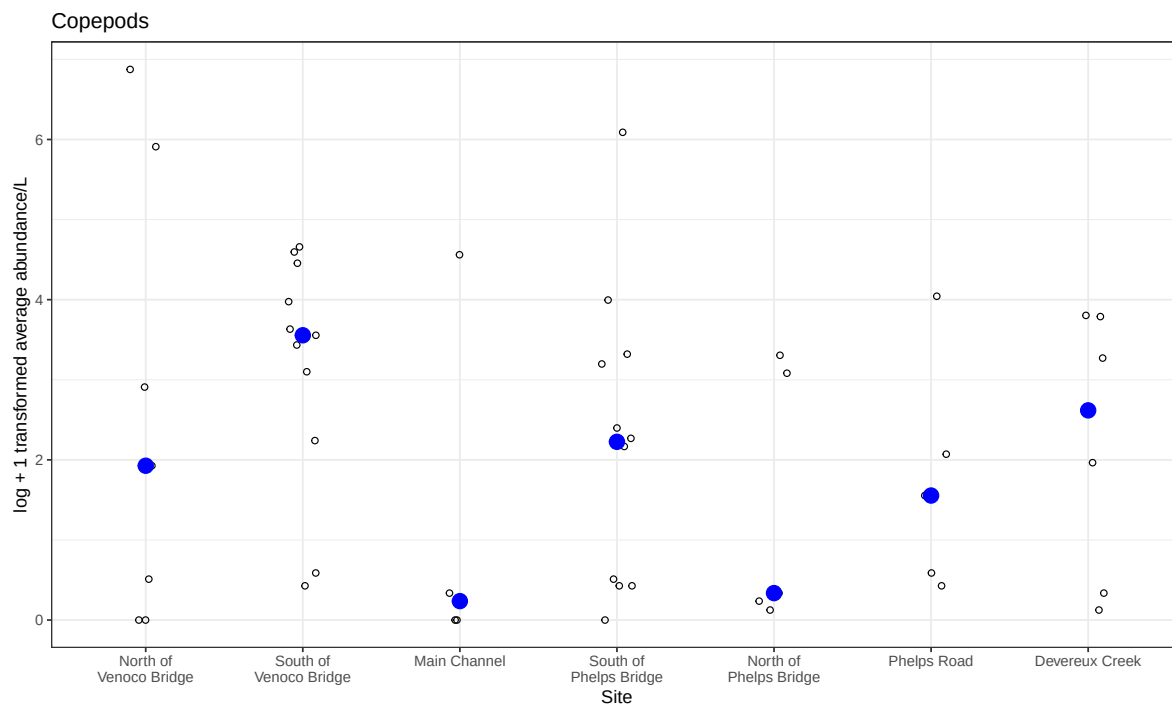
```



```
# base layer
copepods_plot <- ggplot(data = copepods_df,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
```

```
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +
# different theme
theme_bw()

# displaying object
copepods_plot
```



Question 1

A: Planning

- x axis: Site
- y axis: log transformed Ostracod abundance
- geometry abundance: `geom_jitter()`
- geometry median: `stat_summary()`

B: Wrangle Data

```
#new object from aq_ins_clean
ostracod <- aq_ins_clean |>
  # filter to only include ostracod
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
slice_sample(ostracod, n=10)
```

```
# A tibble: 10 x 24
  date_site      date            site taxon ave_lit med_ph med_sal med_do
  <chr>          <dtm>          <chr> <chr>  <dbl>  <dbl>  <dbl>  <dbl>
1 2021-04-15_NPB 2021-04-15 00:00:00 NPB  ostr~  185.    8.09    3.57    8.24
2 2022-05-06_NMC 2022-05-06 00:00:00 NMC  ostr~    0    9.96    1.45   16.5
3 2023-05-05_NPB 2023-05-05 00:00:00 NPB  ostr~  24.7    NA      NA     12.1
4 2023-07-17_NPB 2023-07-17 00:00:00 NPB  ostr~   4.53   NA      NA      NA
5 2023-02-12_NEC 2023-02-12 00:00:00 NEC  ostr~   2.13   NA      NA      NA
6 2022-08-08_NPB2 2022-08-08 00:00:00 NPB2 ostr~    2    7.68    2.36    2.43
7 2022-08-12_NVBR 2022-08-12 00:00:00 NVBR ostr~    0    NA      NA      NA
8 2022-11-12_NEC 2022-11-12 00:00:00 NEC  ostr~    0    7.33    7.01    2.4
9 2024-01-28_NPB 2024-01-28 00:00:00 NPB  ostr~   3.33   7.43    3.08    6.6
10 2022-11-19_NPB1 2022-11-19 00:00:00 NPB1 ostr~    2    7.26    2.47    3.96
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

C: Code Figure

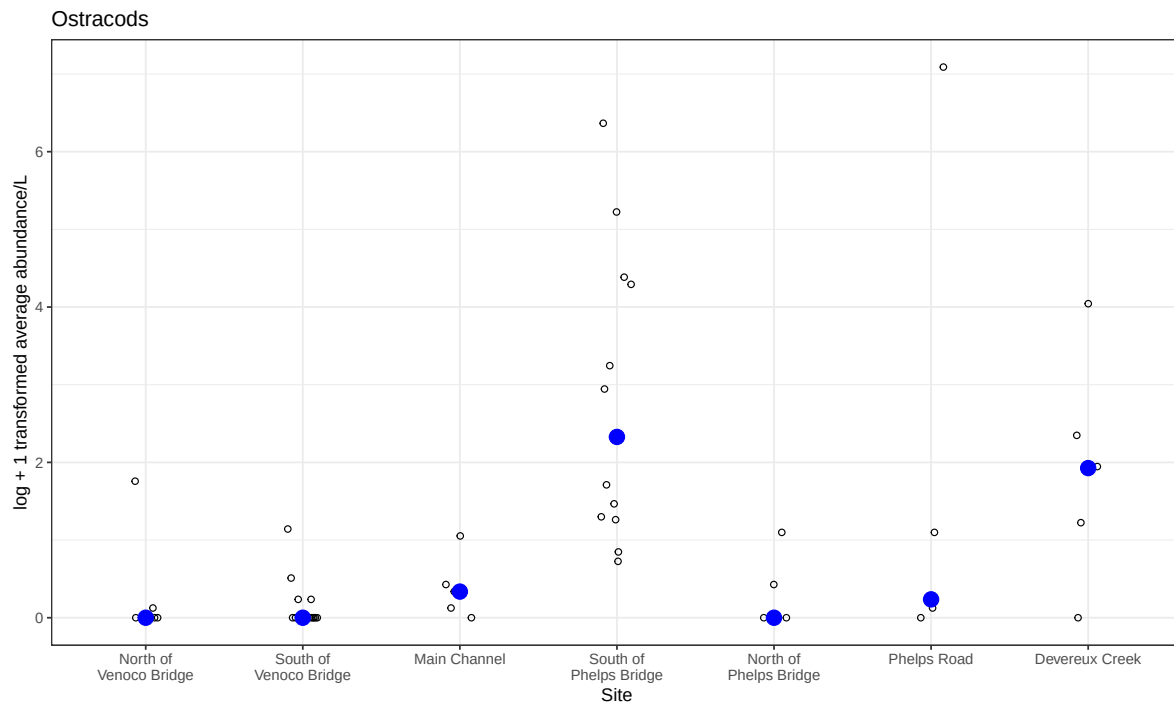
```
# base layer
ostracods_plot <- ggplot(data = ostracod,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
```

```

geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# different theme
theme_bw()

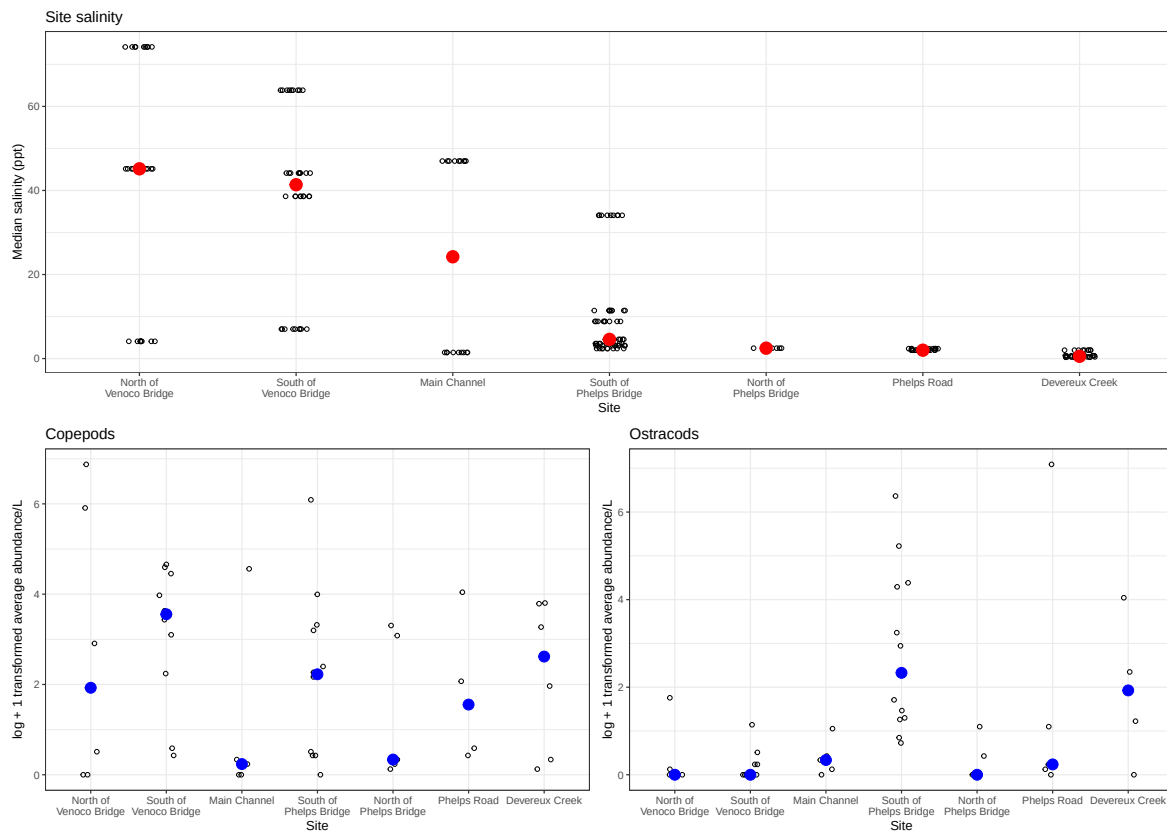
# displaying object
ostracods_plot

```



D: Create multipanel figure

```
#combining plots  
salinity/(copepods_plot+ostracods_plot)
```



E: Analysis

In 1-2 sentences each, describe:

differences between sites in copepod and ostracod abundance how ostracod abundance may relate (or not) to site-level differences in salinity Be specific using the visual components of the plot.

- The differences in sites between copepods and ostracod abundance are mainly at the South of Phelps Bridge as ostracods are the most abundant here while copepods are more abundant South of Venoco Bridge.

- Ostracod abundance may not be related to site-level differences as there are numerous occasions of median salinity being relatively low at places like Phelps Bridge or Phelps road, yet South of Phelps Bridge sites have high abundance while the North side doesn't. Additionally, low abundance sites like North of Phelps have the same abundance as high salinity sites like Venoco Bridge.

Question 2

A: Planning

- x axis: salinity
- y axis: log transformed abundance
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~taxon, scales="free")

B: Wrangle Data

```
# creating new clean object from aquatic inverts
ab_salinity <- aq_ins_clean |>
  # log transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit+1),
    # convert taxon to title
    taxon = to_any_case(taxon, case = "title"),
    # order taxon levels by abundance
    taxon = fct_reorder(taxon, ave_lit, .fun="max"))

#display 10 random rows
slice_sample(ab_salinity, n = 10)
```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>
1	2023-03-03_NPB2	2023-03-03 00:00:00	NPB2	Anne~	0	7.71	1.98	9.59
2	2021-04-15_NPB	2021-04-15 00:00:00	NPB	Clad~	3.33	8.09	3.57	8.24
3	2022-02-23_NVBR	2022-02-23 00:00:00	NVBR	Clad~	0	8.86	45.2	10.2
4	2023-05-05_NPB	2023-05-05 00:00:00	NPB	Dipt~	0	NA	NA	12.1
5	2022-11-18_NVBR	2022-11-18 00:00:00	NVBR	Dipt~	0	NA	NA	NA
6	2022-05-05_NEC	2022-05-05 00:00:00	NEC	Nema~	0	9.45	63.8	7.3


```

7 2023-11-12_NMC 2023-11-12 00:00:00 NMC Anne~ 0 NA 47 14.0
8 2021-11-12_NPB 2021-11-12 00:00:00 NPB Dipt~ 0.133 NA NA NA
9 2021-11-16_NPB 2021-11-16 00:00:00 NPB Nema~ 0 7.3 2.38 12.1
10 2022-05-05_NEC 2022-05-05 00:00:00 NEC Ostr~ 0.267 9.45 63.8 7.3
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
# Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, log_ave_lit <dbl>

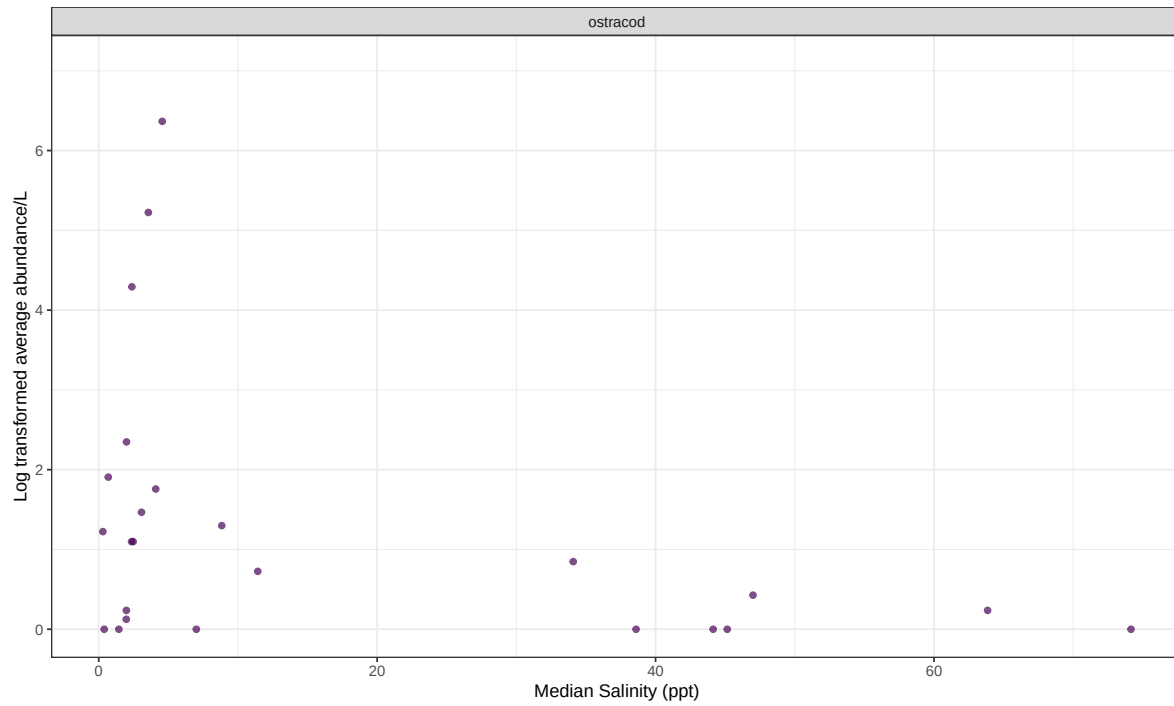
```

C: Code figure

```

# base layer
ggplot(data = ostracod,
       # x-axis: site
       # # y-axis: mean abundance/L
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = taxon)) +
# layer 1: points to represent observations
geom_point(alpha = 0.7) +
#facet wrap for taxon
facet_wrap(~ taxon,
          scales = "free") +
# use color palette
scale_color_viridis_d() +
# labels
labs(x = "Median Salinity (ppt)",
     y = "Log transformed average abundance/L") +
# theme
theme_bw()+
# remove legend from theme
theme(legend.position = "none")

```



D: Analysis

In 3-5 sentences, describe differences or similarities between taxa in relationships between abundance and salinity. Be specific using the visual components of the plot.